Heat-evolved Chebyshev edges and localized fluxes of sparse Fourier shears

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Abstract

Heat evolution of a high-degree Chebyshev polynomial near an endpoint has two logarithmic scales. For $Q_{m,A}(s,c) = (e^{sA^{-2}D^2}T_m)(1+c/A)$ and $\mu = (m^2/A)^{1/3}$, the common amplitude has scale μ^2, while a bounded endpoint displacement has scale μ. I give uniform, nonoptimal error bounds on compact positive time intervals throughout $\mu \to \infty$, $\mu/A \to 0$. The relative-displacement estimate follows from a direct Gaussian tilt, so its error is not obtained by subtracting two coarse amplitude estimates. An edge-profile proposition applies also to the endpoint-normalized Chebyshev polynomials of the second kind.

An explicit sequence of consecutive-integer-frequency shears transfers the first-kind profile over the full time interval, using one initial datum for each member of the sequence. Its signed collar-flux quotient retains an $\exp(\Theta(\mu))$ edge factor with algebraic factors stated separately. The denominator is the cubic mass of the nonconstant component on the whole spatial plateau. For the stated amplitude normalization, inclusion of the constant drift in the denominator instead makes this quotient decay; amplitude rescaling and changes of observation frame are treated explicitly. A supporting time-weighted inequality is a corollary of Erdélyi's results, and an elementary N-term construction shows its $N^{4/3}$ order cannot be reduced in the abstract temporal class. The spatial extrapolation scale and polynomial heat operation are established inputs. The shear application concerns smooth exact solutions and gives no general Navier–Stokes regularity conclusion.

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1 Introduction

A polynomial that is bounded on an interval can grow rapidly just beyond an endpoint. Heat evolution changes both its overall size and the contrast between nearby points. The question studied here is a joint degree–scale limit in which these effects occur at different logarithmic orders. This distinction matters when a common large amplitude cancels from an observation quotient.

For the first Chebyshev family, write

$$Q_{m,A}(s,c) = (e^{sA^{-2}D^2}T_m)(1+c/A), \qquad \mu = (m^2/A)^{1/3}, \qquad \epsilon = \mu/A.$$

Here $D = d/dx$. In the regime $A \to \infty$, $\mu \to \infty$, $\epsilon \to 0$, the limiting normalized amplitude and displacement slope are

$$F_s = 3\,2^{-4/3}s^{1/3}, \qquad G_s = 2^{-2/3}s^{-1/3}.$$

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The quantitative theorem below controls the amplitude error by $C(\epsilon + \log(2+\mu)/\mu^2)$ and the relative-displacement error by a modulus tending to zero with $\sqrt{\epsilon} + \sqrt{\log(2+\mu)}/\mu$. The constants and a sufficient parameter test are given in the proof. No best error order is claimed, and no relation between the rates $\mu \to \infty$ and $\epsilon \to 0$ is added. An $o(\mu^2)$ amplitude remainder alone would not preserve a μ-order displacement effect. The proof instead controls the mean of the Gaussian-tilted edge distribution at each displacement.

Main results. The main result is the quantitative two-scale estimate and its edge-profile formulation, verified for T_m and $U_m/(m+1)$. For the second family the effective degree is $n = m + 1$. The normalization at one removes an otherwise uncontrolled degree factor; at the saddle the remaining prefactor has logarithmic size $O(\log \mu)$. The two families therefore test a substantive hypothesis of the proposition, rather than merely a change of polynomial notation.

The application uses one fixed initial profile for each member of a sequence, throughout $0 \le t \le 4R^2$. An exact finite Fourier shear approximates the polynomial heat evolution uniformly on this interval. The flux proof keeps the opposed-cap signs, the early time contribution, a terminal interval of polynomial width, and the full three-dimensional plateau, including its tapering ends. A profile selected afresh at each time would not give this statement.

Relation to prior work. The finite heat operation on polynomial spaces is classical; Hile and Stanoyevitch's heat-polynomial framework and Hermite–Chebyshev operational formulas are relevant precedents. Their precise relation to the present positive heat time, varying degree, and displaced edge is recorded in Section 3. The terminating hypergeometric representation also explains the origin of the leading constant through a classical ${}_0F_2$ model. A fixed parameter large-argument expansion does not by itself control the terminating degree-dependent series or its smaller displacement scale. I do not claim a new special-function object or a new Laplace method. The literature comparison is bounded and does not establish priority.

The sparse-Fourier scale $N \exp(CN\sqrt{d})$ and the basic Chebyshev and endpoint witnesses are due to established work, particularly Zhang's preprint [10] and the inequalities collected by Erdélyi [2]. They are compressed into an attributed input and a short appendix. The auxiliary time-weighted upper estimate is likewise a direct literature corollary. Its explicit order witness completes the abstract temporal comparison; it does not make the spatially constrained flux upper bound optimal.

Observation frame and normalization. The exact shear $u = (0, B, F(t, x_2))$ satisfies $F_t + BF_{x_2} = F_{x_2x_2}$. A Galilean change $y = x_2 - Bt$ and subtraction of the constant velocity yield a pure heat shear $(0, 0, H(t, y))$. For a radial window fixed in this new frame, the remaining cubic transport term integrates to zero by symmetry. The original fixed-window flux is therefore frame dependent: it measures constant transport through that window, without introducing vortex stretching or a new nonlinear singularity mechanism.

The mass in the main application is $\int |F|^3$ over the full spatial plateau and clock. The word "full" describes the region, not all velocity components. In the original-amplitude family, replacing it by $\int (B^2 + F^2)^{3/2}$ yields a vanishing quotient with quadratic logarithmic rate $-9/20000$. This conclusion depends on normalization: rescaling the nonconstant initial profile sufficiently restores the component quotient, while retaining large drift and unbounded energy across the sequence. Section 5 states both facts precisely. The application is an explicit model of observation and normalization, not an obstruction for general finite-energy solutions.

The quantitative edge theorem comes first, followed by its fixed-data shear realization and frame analysis. The auxiliary uniform flux upper bound follows. Appendices give the temporal order comparison, classical spatial witnesses, normalization consequences, and a finite diagnostic.

2 Quantitative Gaussian edge estimates

The following estimates quantify the two scales without a relation between their two small errors. The proof uses the mean under each displaced Gaussian measure. In particular, it does not divide two estimates whose errors are only small on the common-amplitude scale.

Theorem 2.1 (Uniform errors for two Chebyshev families)**.** *Fix $0 < s_0 \le s_1 < \infty$ and $0 \le C_0 < \infty$. Consider either*

$$p_n = T_n \quad (n \ge 1), \qquad \text{or} \qquad p_n = U_{n-1}/n \quad (n \ge 1),$$

where $U_{n-1}(\cos\theta) = \sin(n\theta)/\sin\theta$. Put

$$Q^p_{n,A}(s,c) = (e^{sA^{-2}D^2}p_n)(1 + c/A), \quad \mu = (n^2/A)^{1/3}, \quad \epsilon = \mu/A,$$

and set $F_s = 3\,2^{-4/3}s^{1/3}$ and $G_s = 2^{-2/3}s^{-1/3}$. There are constants $C, \mu_* > 0$ and $\epsilon_* > 0$, computable from s_0, s_1, C_0 alone, such that if $\mu \ge \mu_*$ and $0 < \epsilon \le \epsilon_*$, then $Q^p_{n,A}(s,c) > 0$ and

$$\left|\mu^{-2}\log Q^p_{n,A}(s,0) - F_s\right| \le C\left(\epsilon + \frac{\log(2+\mu)}{\mu^2}\right), \tag{1}$$

$$\left|\mu^{-1}\log\frac{Q^p_{n,A}(s,c_2)}{Q^p_{n,A}(s,c_1)} - G_s(c_2 - c_1)\right| \le C|c_2 - c_1|\left(\sqrt{\epsilon} + \frac{\sqrt{\log(2+\mu)}}{\mu}\right). \tag{2}$$

These estimates hold uniformly for $s \in [s_0, s_1]$ and $|c_1|, |c_2| \le C_0$. No optimality of these error orders is asserted. In particular, they apply throughout $A \to \infty$, $\mu \to \infty$, $\mu/A \to 0$.

For auditability, a profile proposition gives explicit constants before they are absorbed into C.

Proposition 2.2 (A square-root edge profile criterion)**.** *Write $\psi(z) = \sqrt{2z}$, $z_- = 2^{1/3}s_0^{2/3}$, $z_+ = 2^{1/3}s_1^{2/3}$, $Z = z_+ + 1$, and $J = [z_-, Z]$. Suppose a real measurable profile $p = p_{\mu,A}$, with $A = \mu/\epsilon$, satisfies the following inequalities for some fixed $k > 0$, $b, d \ge 0$:*

$$|p(x)| \le 1 \qquad (|x| \le 1), \tag{3}$$

$$0 < p(1 + \mu z/A) \le e^{\mu^2\psi(z)}, \quad |p(-1 - \mu z/A)| \le e^{\mu^2\psi(z)} \qquad (z \ge 0), \tag{4}$$

$$p(1 + \mu z/A) \ge k\mu^{-b}e^{\mu^2(\psi(z) - d\epsilon)} \qquad (z \in J). \tag{5}$$

Assume these hold for $\mu \ge \mu_0 \ge 1$ and $0 < \epsilon \le \epsilon_0 \le 1$. Define Q^p by its Gaussian convolution, which agrees with the finite heat expansion for a polynomial:

$$Q^p(s,c) = \frac{1}{\sqrt{4\pi s}}\int_{\mathbb{R}} p(1 + y/A)e^{-(y-c)^2/(4s)}\,dy.$$

The conclusions of Theorem 2.1 hold. Constants may also depend on $k, b, d, \mu_0, \epsilon_0$. The formulas below specify the constants and a noncircular threshold.

Constants for the profile criterion. In the following formulas, $\log_+ x = \max\{0, \log x\}$. All constants are fixed before μ, A are chosen. Set

$$F_* = 3\,2^{-4/3}s_0^{1/3}, \qquad D_* = (2z_-)^{-3/2} + (2s_0)^{-1}, \qquad a_* = (8s_1)^{-1},$$

$$H_* = C_0^2 s_1/(2s_0^2), \qquad L_* = \frac{k}{\sqrt{4\pi s_1}}e^{-D_*/2 - C_0/(2s_0)}, \qquad V_* = \sqrt{2s_1/s_0}\,e^{H_*},$$

$$K_* = \frac{e^{H_*}}{\sqrt{4\pi s_0}L_*}, \qquad H_{\mathrm{tail}} = 3\,2^{-4/3}(2s_1)^{1/3}, \qquad E_0 = 1 + \sqrt{s_1/(2s_0)},$$

$$E_1 = \frac{1}{\sqrt{\pi s_0}} + \frac{6\sqrt{s_1/e}\sqrt{2\pi s_1} + 4s_1}{2s_0\sqrt{4\pi s_0}}, \qquad \kappa = F_*/4.$$

For $\mu \geq 1$ put

$$\begin{aligned} r(\mu,\epsilon)^2 &= \frac{d\epsilon}{a_*} + \frac{(b+1)\log\mu + \log_+(K_*/a_*)}{a_*\mu^2}, \\ \rho(\mu,\epsilon) &= r(\mu,\epsilon) + \mu^{-2}, \\ \rho_* &= \sqrt{\frac{d+b+1+\log_+(K_*/a_*)}{a_*}} + 1, \\ M_* &= (z_+ + \rho_* + C_0)/(2s_0), \qquad E_2 = 2(E_1 + E_0 M_*). \end{aligned}$$

One admissible threshold is

$$\epsilon_* = \min\left\{\epsilon_0,\ \frac{1}{\sqrt{8s_1 H_{\mathrm{tail}}}},\ \frac{F_*}{4d}\right\}, \tag{6}$$

$$\mu_* = \max\left\{2, C_0, \mu_0, \frac{2C_0 z_+}{s_0 F_*}, \frac{8b}{F_*}, \sqrt{\frac{8\{\log_+(1/L_*) + C_0^2/(4s_0)\}}{F_*}}, \sqrt{\frac{\log(2E_0)}{\kappa}}\right\}. \tag{7}$$

The entry $F_*/(4d)$ is omitted when $d = 0$. At this threshold, more explicit versions of the two errors are

$$\left|\mu^{-2}\log Q^p(s,0) - F_s\right| \leq d\epsilon + \frac{b\log\mu + \max\{\log_+(1/L_*), \log_+ V_*\} + 2E_0 e^{-\kappa\mu^2}}{\mu^2}, \tag{8}$$

$$\left|\mu^{-1}\log\frac{Q^p(s,c_2)}{Q^p(s,c_1)} - G_s(c_2-c_1)\right| \leq |c_2-c_1|\left\{\frac{\rho(\mu,\epsilon) + C_0/\mu}{2s_0} + E_2 e^{-\kappa\mu^2}\right\}. \tag{9}$$

Proof of Proposition 2.2. Retain the positive exterior and write

$$I(s,c) = \frac{1}{\sqrt{4\pi s}}\int_0^\infty p(1+y/A)e^{-(y-c)^2/(4s)}\,dy.$$

Let $z_s = 2^{1/3}s^{2/3}$ and $\Phi_s(z) = \psi(z) - z^2/(4s)$. Its derivative vanishes at z_s, $\Phi_s(z_s) = F_s$, and

$$\Phi_s''(z) = -(2z)^{-3/2} - (2s)^{-1} \leq -(2s_1)^{-1} \quad (z > 0).$$

Consequently, including $z = 0$ by continuity,

$$\Phi_s(z) \leq F_s - \frac{(z-z_s)^2}{4s_1} \quad (z \geq 0). \tag{10}$$

Put $\mathcal{B} = \mu^2 F_s + c\mu z_s/(2s) - c^2/(4s)$. On $z_s \leq z \leq z_s + 1/\mu$, the interval lies in J, $|\Phi_s''(z)| \leq D_*$, and the linear tilt changes by at most $C_0/(2s_0)$. Integrating the local lower bound over this interval therefore gives

$$I(s,c) \geq L_*\mu^{-b}e^{\mathcal{B}-d\mu^2\epsilon}. \tag{11}$$

For the upper bound, Young's inequality in the precise form

$$\frac{c\mu(z-z_s)}{2s} \leq \frac{\mu^2(z-z_s)^2}{8s_1} + H_*$$

and (10) show that

$$I(s,c) \leq V_* e^{\mathcal{B}}. \tag{12}$$

Indeed the changed measure is $\mu\,dz/\sqrt{4\pi s}$ and its Gaussian upper bound can be integrated over the whole real line.

Normalize the positive integrand for each c to a probability measure for $Z = y/\mu$. Equations (11) and (10) give its density bound

$$f_{s,c}(z) \le K_* \mu^{b+1} e^{d\mu^2\epsilon} e^{-a_*\mu^2(z-z_s)^2} \quad (z \ge 0).$$

For any $r \ge 0$, extending both tails to the whole line gives

$$\mathbb{E}_{s,c}|Z - z_s| \le r + \frac{K_*}{a_*}\mu^{b-1} e^{d\mu^2\epsilon - a_*\mu^2 r^2}.$$

The displayed choice of $r(\mu,\epsilon)$ makes the second term at most μ^{-2}. Differentiating the Gaussian, justified by the global envelope, now yields the exact identity

$$\partial_c \log I(s,c) = \frac{\mu \mathbb{E}_{s,c} Z - c}{2s}.$$

It follows that

$$\left|\mu^{-1}\partial_c \log I(s,c) - G_s\right| \le \frac{\rho(\mu,\epsilon) + C_0/\mu}{2s_0}. \tag{13}$$

This estimate treats the unbounded Gaussian tilt directly and uniformly for positive and negative c.

It remains to restore the signed remainder $\mathcal{R} = Q^p - I$. Its interior contribution from $-2A \le y \le 0$ has absolute value at most one and c-derivative at most $(\pi s_0)^{-1/2}$. In the negative exterior put $y = -2A - u$, $u \ge 0$. Since $A \ge C_0$,

$$2A + u + c \ge A + u, \qquad \mu^2\psi(u/\mu) - \frac{(2A+u+c)^2}{4s} \le -\frac{A^2}{4s_1} + H_{\text{tail}}\mu^2 - \frac{u^2}{8s_1}.$$

The value of H_{tail} follows by maximizing $\sqrt{2z} - z^2/(8s_1)$ on $z \ge 0$; its maximizer is $2^{1/3}(2s_1)^{2/3}$. The threshold for ϵ makes the right side at most $-A^2/(8s_1) - u^2/(8s_1)$. Gaussian integration proves $|\mathcal{R}| \le E_0$. For its derivative use $|2A+u+c|/(2s) \le (3A+u)/(2s_0)$, together with

$$\int_0^\infty e^{-u^2/(8s_1)}du = \sqrt{2\pi s_1}, \quad \int_0^\infty u e^{-u^2/(8s_1)}du = 4s_1, \quad \sup_{A\ge 0} A e^{-A^2/(8s_1)} = 2\sqrt{s_1/e}.$$

This proves $|\partial_c \mathcal{R}| \le E_1$ with the stated constant. No assumption on the sign of either omitted portion is used.

The first threshold gives $d\epsilon \le F_*/4$. The linear term in $\mathcal{B}$ costs at most $F_*\mu^2/4$ by the fourth entry of (7). Finally $\log\mu \le \mu$, the fifth and sixth entries, and (11) give

$$I(s,c) \ge e^{\kappa\mu^2}, \qquad |\mathcal{R}/I| \le E_0 e^{-\kappa\mu^2} \le \tfrac{1}{2}.$$

Thus $Q^p > 0$ and $|\log Q^p - \log I| \le 2E_0 e^{-\kappa\mu^2}$. Since $\epsilon \le 1$ and $\log\mu \le \mu^2$, we have $\rho \le \rho_*$ and $|\partial_c \log I| \le \mu M_*$. Hence

$$\mu^{-1}|\partial_c \log Q^p - \partial_c \log I| = \mu^{-1}\left|\frac{\partial_c\mathcal{R} - \mathcal{R}\partial_c \log I}{I + \mathcal{R}}\right| \le E_2 e^{-\kappa\mu^2}.$$

Integrating this estimate and (13) between c_1 and c_2 proves (9), including exact vanishing when $c_1 = c_2$. At $c = 0$, use (11) and (12) to obtain (8). These explicit estimates imply the simpler displayed orders in the theorem. For example, $e^{-\kappa\mu^2} \le (2e\kappa)^{-1/2}/\mu$ for $\mu > 0$, and each remaining constant term is absorbed using $\log(2+\mu) \ge 1$ for $\mu \ge 2$. □

Verification of Theorem 2.1. Write $\theta = \operatorname{arcosh}(1+\epsilon z)$. For $u \geq 0$, integrating $(2t)^{-1/2}(1 + t/2)^{-1/2}$ from zero to u gives

$$\sqrt{2u}(1 - u/12) \leq \operatorname{arcosh}(1+u) \leq \sqrt{2u}, \qquad \operatorname{arcosh}(1+u) \geq \frac{\sqrt{2u}}{\sqrt{1+u/2}}. \tag{14}$$

The first lower bound uses $(1+v)^{-1/2} \geq 1 - v/2$; the other uses the minimum of the decreasing factor in the same integral. As $n\sqrt{\epsilon} = \mu^2$, it follows on J that

$$\mu^2\{\psi(z) - d\epsilon\} \leq n\theta \leq \mu^2\psi(z), \qquad d = \sqrt{2}Z^{3/2}/12.$$

For T_n, the identity $T_n(\cosh\theta) = \cosh(n\theta)$ gives the exterior envelope and local lower bound with $k = 1/2$, $b = 0$, $\mu_0 = 1$, $\epsilon_0 = 1$. Its interior bound and parity give the other two hypotheses.

For the second family, the finite sum

$$\frac{U_{n-1}(\cosh\theta)}{n} = \frac{1}{n}\sum_{j=0}^{n-1} e^{(n-1-2j)\theta} = \frac{\sinh(n\theta)}{n\sinh\theta}$$

is positive and at most $e^{n\theta}$. The corresponding trigonometric sum has modulus at most one, proving the interior bound, including the endpoints by continuity. Parity gives the negative exterior envelope. On J one has the exact denominator identity

$$n\sinh\theta = \mu^2\sqrt{z(2+\epsilon z)} \leq \mu^2\sqrt{Z(2+Z)}.$$

Put $h_* = \sqrt{2z_-}/\sqrt{1+Z/2}$ and $\mu_0 = \max\{1, h_*^{-1/2}\}$. Then $n\theta \geq \mu^2 h_* \geq 1$ on J by (14). Therefore

$$\frac{U_{n-1}(1+\epsilon z)}{n} \geq \frac{1-e^{-2}}{2\sqrt{Z(2+Z)}}\mu^{-2}e^{\mu^2\{\psi(z)-d\epsilon\}}.$$

This verifies the criterion with $b = 2$, $k = (1-e^{-2})/(2\sqrt{Z(2+Z)})$, and $\epsilon_0 = 1$. All constants and thresholds are consequently obtained from the explicit formulas above. Taking their maximum or minimum as appropriate gives one choice valid for both families. □

Remark 2.3 (Normalization and the scope of the extension)**.** For unnormalized U_{n-1} the displacement ratio is unchanged, but the common-amplitude formula contains the additional term $\mu^{-2}\log n$. This term need not vanish in the original joint range. For example, take $n_j = \lceil e^{j^3} \rceil$ and $A_j = n_j^2/j^3$. Then $\mu_j = j$, $\epsilon_j = j^4/n_j^2 \to 0$, whereas $\mu_j^{-2}\log n_j \sim j$. The normalization therefore removes a real obstruction, rather than only changing notation. The profile criterion requires positivity on the exterior, a global envelope on both sides, an interior bound, and a quantitative lower bound near the saddle. It does not assert the same conclusion for arbitrary Jacobi parameters or arbitrary polynomial heat flows. The normalized second family has the same two leading constants; no Navier–Stokes realization for that additional family is claimed here.

3 Special-function context and the contribution boundary

Polynomial heat evolution and its finite operational expansion are classical. Hile and Stanoyevitch's framework [4, Theorem 1 and (2.18)] includes polynomial initial data for constant-coefficient evolution equations. Batahan and Shehata [1, Theorem 3.1] give a Hermite–Chebyshev operational representation with a fixed negative heat parameter and the second-kind polynomial. Related two-variable formulas appear in Khan [5]. These formulas are cited as historical context, not as inputs to the positive-time estimates above; the finite Gaussian representation and the two normalizations used here are verified directly.

The standard trigonometric definitions of the two Chebyshev families are recorded in [7, §18.5]. At zero displacement there is also an exact terminating identity:

$$Q_{m,A}(s,0) = {}_4F_2\left(\begin{matrix}-m/2,(1-m)/2,m/2,(m+1)/2\\ 1/4,3/4\end{matrix};\frac{s}{A^2}\right). \tag{15}$$

Here the notation means the finite sum through $j = \lfloor m/2 \rfloor$. To verify it, let $D_k = T_m^{(k)}(1)$ and differentiate the Chebyshev ODE at one. The recurrence $D_{k+1}/D_k = (m^2-k^2)/(2k+1)$, $D_0 = 1$, shows that the ratio of successive terms $D_{2j}(s/A^2)^j/j!$ is exactly the hypergeometric term ratio in (15). The constant terms also agree. This is an algebraic representation, not an additional asymptotic theorem.

Its leading coefficient model is

$${}_0F_2(;1/4,3/4;s\mu^6/16).$$

The classical exponential factor $\exp(3z^{1/3})$ of this function explains $F_s\mu^2$; see [7, §16.11]. It does not imply that the two functions have ratio tending to one. Indeed the exact jth coefficient relative to that model contains

$$\prod_{k=0}^{2j-1}(1-k^2/m^2).$$

For $j \asymp \mu^2$ in the original regime, $j/m \to 0$, and the logarithmic correction has order $j^3/m^2 \asymp \mu^3/A$. This follows by summing $-2x \le \log(1-x) \le -x$ once $(2j/m)^2 \le 1/2$. That correction is $o(\mu^2)$ but need not be $o(1)$.

The large-variable expansions in DLMF §16.11(ii) explicitly exclude nonpositive integer upper parameters and treat $p \le q+1$. They therefore do not directly apply to this terminating $p = q+2$ family. The large-parameter formulas in §16.11(iii) use other parameter paths. The uniform Gauss hypergeometric expansions of Paris [8, 9] concern ${}_2F_1$ with specified large-parameter paths and transition regions. They do not state the terminating four-upper-parameter heat-edge estimate and its bounded spatial-displacement quotient used here. This comparison does not rule out other covering results.

Large-degree polynomial heat flow also appears in work on zeros and random matrices, for example Hall and Ho [3]. A macroscopic zero distribution differs from the function values and relative amplitudes at shrinking endpoint distances considered here. The bounded primary-source search found no theorem with the same joint regime, explicit uniform errors, and displacement quantifiers. It is not an exhaustive search of all terminating hypergeometric literature and establishes no first-result or priority claim.

The claimed addition is the quantitative statement with these hypotheses, its checked second-family instance, and the fixed-data full-interval realization below. The Laplace principle, Gaussian tilting, and polynomial heat operation remain established methods.

4 A fixed-initial-data realization by integer Fourier shears

The lower bounds at a prescribed time do not determine what happens over a time interval. This section treats one explicit family with its coefficients specified at time zero. The heat operator has a positive time parameter throughout.

Let T_m be the Chebyshev polynomial of the first kind, defined by $T_m(\cos\theta) = \cos(m\theta)$. For $A > 0$ set

$$Q_{m,A}(s,c) = \left(e^{sA^{-2}D^2}T_m\right)(1+c/A), \qquad \mu = (m^2/A)^{1/3}. \tag{16}$$

The exponential is a finite sum when applied to a polynomial. All limits in this section are joint limits satisfying

$$A \longrightarrow \infty, \qquad m/\sqrt{A} \longrightarrow \infty, \qquad m/A^2 \longrightarrow 0. \tag{17}$$

Equivalently, $\mu \to \infty$ and $\mu/A \to 0$. No convergence rate in these two conditions is imposed.

Theorem 4.1 (Two scales in the heat-evolved edge)**.** *Under* (17)*, for every compact interval* $S \subset (0,\infty)$ *and every* $C_0 < \infty$*,* $Q_{m,A}(s,c)$ *is positive for all sufficiently large parameters, uniformly for* $s \in S$ *and* $|c| \le C_0$*. Moreover,*

$$\mu^{-2} \log Q_{m,A}(s,0) \longrightarrow F_s := 3\,2^{-4/3} s^{1/3}, \tag{18}$$

$$\mu^{-1} \log \frac{Q_{m,A}(s,c_2)}{Q_{m,A}(s,c_1)} \longrightarrow G_s(c_2 - c_1), \qquad G_s := 2^{-2/3} s^{-1/3}. \tag{19}$$

Both convergences are uniform for $s \in S$ *and* $|c_1|, |c_2| \le C_0$*. In particular* $F_4 = 3\,2^{-2/3}$ *and* $G_4 = 2^{-4/3}$*.*

Proof. This is the first-kind case of Theorem 2.1. Its sufficient parameter test holds eventually along every sequence in (17); both stated error bounds tend to zero. In particular the displacement limit uses the separate Gaussian-tilt estimate, not subtraction of amplitude estimates. □

Lemma 4.2 (A majorant and a terminal time interval)**.** *Put* $D_k = T_m^{(k)}(1)$*. Then*

$$Q_{m,A}(s,c) = \sum_{\ell+2j \le m} \frac{D_{\ell+2j}}{A^{\ell+2j}\ell! j!} c^\ell s^j, \qquad \frac{D_{k+1}}{D_k} = \frac{m^2 - k^2}{2k+1} \quad (0 \le k < m). \tag{20}$$

All D_k *in this range are positive. For* s *in a fixed compact subinterval of* $(0,\infty)$ *and* $0 < c_0 \le c \le C_0$*,*

$$0 \le \partial_s \log Q \le C(1+\mu^2), \qquad 0 \le \partial_c \log Q \le C(1+\mu). \tag{21}$$

Thus changes of size at most $h_\mu = (1+\mu^2)^{-1}$ *in* s *and* c *change* Q *by at most a fixed factor on these compact sets. If* $d \ge 0$*,* $0 \le s \le S_1$*, and* $|x| \le 1 + d/A$*, then*

$$\left| \left(e^{sA^{-2}D^2} T_m \right)(x) \right| \le Q_{m,A}(S_1, d). \tag{22}$$

Proof. Differentiate $(1-x^2)T_m'' - xT_m' + m^2 T_m = 0$ k times and evaluate at $x = 1$. This gives the displayed recurrence, starting from $D_0 = 1$. The finite Taylor expansion followed by the finite heat expansion gives (20). At fixed ℓ, the ratio of the $(j+1)$st and jth nonzero summands is at most

$$\frac{sm^4}{A^2(j+1)(4j+1)(4j+3)} \le \frac{C\mu^6}{(j+1)^3}.$$

For $j \ge J = \lceil C_1(1+\mu^2) \rceil$ this is at most $1/2$. Under the normalized weights in that row, the tail above J is bounded by a geometric sequence, so its mean index is at most $J+2$. Rows ending before J satisfy the same bound, and zero rows are ignored. Averaging over rows gives $\partial_s \log Q = \mathbb{E}(j)/s \le C(1+\mu^2)$. The identity $\partial_c^2 Q = \partial_s Q$ yields $\mathbb{E}[\ell(\ell-1)] = c^2 \partial_s Q/Q$. Since $\ell \le 1 + \sqrt{\ell(\ell-1)}$, Cauchy–Schwarz gives $\partial_c \log Q \le c^{-1} + \sqrt{\partial_s Q/Q}$. Integrating these derivative estimates proves the assertion about h_μ.

For completeness, $|T_m^{(k)}(x)| \le D_k$ for $|x| \le 1$. For $k \ge 1$ this follows from $T_m^{(k)} = m 2^{k-1}(k-1)! C_{m-k}^k$: the generating function

$$\sum_{n \ge 0} C_n^k(\cos\theta) t^n = (1 - te^{i\theta})^{-k}(1 - te^{-i\theta})^{-k}$$

has nonnegative absolute coefficient majorants given by its value at $\theta = 0$. The case $k = 0$ is the defining Chebyshev bound. For $1 \le |x| \le 1 + d/A$, parity and the positive Taylor coefficients at one instead give $|T_m^{(k)}(x)| \le T_m^{(k)}(1 + d/A)$. The heat series has nonnegative coefficients in s, so termwise use of these derivative majorants proves (22). □

4.1 Uniform transfer to consecutive integer modes

The following estimate keeps the polynomial degree in the error bound. This is necessary because pointwise convergence at fixed degree would not justify the subsequent limit.

Proposition 4.3 (Dynamic confluence)**.** *Fix $\delta_0, \beta > 0$ and let*

$$a = \frac{32}{63}L, \quad A = a - \delta_0, \quad R = e^{-9L^2/40000}, \quad e_a = A/a, \quad v = -\beta/A, \quad B = av/R. \tag{23}$$

Suppose m is a positive even integer satisfying (17). Set $\eta = AR$ and

$$w_\eta(x) = \frac{e^{i\eta x} - 1}{i\eta}, \qquad T_m(w_\eta(x)) = \sum_{j=0}^{m} b_j(\eta) e^{ij\eta x}, \qquad b_j(\eta) = \frac{T_m^{(j)}(i/\eta)}{j!(i\eta)^j}. \tag{24}$$

Every b_j is nonzero. Define the real periodic solution

$$F(t,y) = 2\,\mathrm{Re} \sum_{j=0}^{m} b_j(\eta) e^{-(m+j)^2 t + i(m+j)(y - Bt)}, \qquad u(t,x) = (0, B, F(t, x_2)). \tag{25}$$

Then u is a smooth unforced unit-viscosity Navier–Stokes solution with constant pressure on $(\mathbb{R}/2\pi\mathbb{Z})^3$ and exactly $m+1$ positive frequencies $m, \dots, 2m$ in its third component. Writing $G(s,z) = F(R^2 s, aRz)$ and $P_s(x) = (e^{sA^{-2}D^2} T_m)(x)$, one has

$$\sup_{0 \le s \le 4,\ |x| \le 2} |G(s, e_a x) - 2P_s(x - vs/e_a)| \le \exp\left(-\frac{9}{40000} L^2 + o(L^2) \right). \tag{26}$$

Proof. The coefficient identity is the Taylor formula for T_m at i/η. Every derivative of a real-rooted polynomial is real-rooted, by Rolle's theorem, and T_m has only real roots. Thus $T_m^{(j)}(i/\eta) \neq 0$ for $j < m$; the derivative of order m is a nonzero constant. The equation $(\partial_t + B\partial_y - \partial_y^2)F = 0$ follows directly from the finite sum. It gives the claimed divergence-free shear solution.

Put $M = m\eta$ and $t_s = s/A^2$. The complex half of $G(s, e_a x)$ is exactly

$$H(s,x) = e^{-M^2 t_s + iM(x - vs/e_a)} \left(e^{t_s D^2} T_m(w_\eta(\cdot)) \right)(x - vs/e_a + 2iMt_s). \tag{27}$$

This identity can be checked on every exponential in (24); in particular the imaginary shift has a positive sign. Under the change of variable $w = w_\eta(x)$, the exact polynomial generator is

$$\mathcal{L}_\eta = (1 + i\eta w)^2 D_w^2 + i\eta(1 + i\eta w) D_w.$$

It maps polynomials of degree at most m to the same space. The chain rule gives, for every polynomial p,

$$D_x^2(p \circ w_\eta) = (\mathcal{L}_\eta p) \circ w_\eta, \qquad e^{tD_x^2}(p \circ w_\eta) = (e^{t\mathcal{L}_\eta} p) \circ w_\eta.$$

The second identity follows from the first by the exponential series: both operators act on the same finite-dimensional invariant space, and the matrix series converge at every finite time. In the norm $\|p\|_4 = \sum_{k=0}^m |p_k| 4^k$ on that space,

$$\|\mathcal{L}_\eta - D_w^2\| \le C\eta m^2, \qquad \|\mathcal{L}_\eta\| + \|D_w^2\| \le Cm^2$$

for $0 < \eta \le 1$. These bounds follow termwise from $\|D_w\| \le m/4$ and $\|D_w^2\| \le m^2/16$. The recurrence $T_{m+1} = 2wT_m - T_{m-1}$ gives $\|T_m\|_4 \le C_4^m$. Duhamel's formula therefore gives

$$\|e^{t_s \mathcal{L}_\eta} T_m - e^{t_s D_w^2} T_m\|_4 \le C t_s \eta m^2 C_4^m e^{C t_s m^2}.$$

For $|x| \le 2$, all the complex arguments in (27) have modulus less than $5/2$ for large L, and their w_η images have modulus less than 3. Thus they remain inside the coefficient disk of radius four. On these smaller disks $|w_\eta(z) - z| \le C\eta$, while one derivative of a polynomial costs at most $m/4$ in the coefficient norm. Replacing the generator, replacing $w_\eta(z)$ by z, removing the imaginary shift $2iMt_s$, and removing the carrier and scalar factor in (27) consequently give a total error at most

$$\eta \ \mathrm{poly}(m, A) \exp(Cm + Cm^2/A^2).$$

The real drift vs/e_a has been retained in the comparison function. Now $\log \eta = -9L^2/40000 + O(\log L)$, $m = o(A^2) = o(L^2)$, and $m^2/A^2 = o(L^2)$. Taking twice the real part proves (26). □

Remark 4.4. Every fixed member of (25) has finite energy on the torus. The coefficients and the constant drift need not remain bounded as $L \to \infty$. The periodic extension to $\mathbb{R}^3$ has infinite total energy and is used only in local coordinates. No uniform energy class or limiting weak solution is asserted here.

4.2 The signed flux and the whole spatial plateau

Fix $0 < \delta_0 < \delta$ and a nonnegative smooth function ϑ supported in $(-\delta, \delta)$ that equals one on $[-\delta_0, \delta_0]$. Suppose there are $r_c, h, \vartheta_* > 0$ such that

$$\delta_0 < r_c - 2h < r_c + 2h < \delta, \qquad \vartheta(r) \ge \vartheta_* \quad \text{for } |r - r_c| \le 2h. \tag{28}$$

Let $\chi \in C^1([0,4])$ satisfy $0 \le \chi \le 1$, $\chi(0) = 0$, and $\chi = 1$ on a fixed neighborhood of four. Define the annular cutoff and its plateau by

$$\phi_L(x) = \vartheta(|x|/R - a), \qquad \mathcal{P}_L = \{x : (a - \delta_0)R \le |x| \le (a + \delta_0)R\}.$$

For large L these sets lie inside a Euclidean coordinate neighborhood of the torus. For the shear of Proposition 4.3 put

$$\mathcal{T}_L = \frac{1}{2} \int_0^{4R^2} \chi(t/R^2) \int_{\mathbb{T}^3} |u(t,x)|^2 u(t,x) \cdot \nabla \phi_L(x) \, dx \, dt, \tag{29}$$

$$M_L^{\mathrm{plat}} = \int_0^{4R^2} \int_{\mathcal{P}_L} |F(t, x_2)|^3 \, dx \, dt. \tag{30}$$

The denominator integrates the oscillatory velocity component over the entire spatial plateau and the entire time interval. It is not the L^3 mass of the full velocity: the constant drift B is not included in (30).

Theorem 4.5 (A full-interval residual for one exact shear family)**.** *Use the parameters and shear of Proposition 4.3 and the fixed cutoffs above. Write*

$$c_- = r_c - h + \delta_0, \qquad c_p = 2\delta_0, \qquad \Delta_c = c_- - c_p = r_c - h - \delta_0 > 0.$$

There are constants $c, C > 0$ depending only on the fixed cutoffs and β such that, along every degree sequence (17), eventually $\mathcal{T}_L > 0$ and

$$\frac{c\beta a^{2/3}}{A(1+\mu^2)} \exp\Big((2^{-1/3}\Delta_c + o(1))\mu\Big) \le R^{1/3} \frac{\mathcal{T}_L}{(M_L^{\mathrm{plat}})^{2/3}} \tag{31}$$

$$\le \frac{C\beta a^{4/3}}{A} (1+\mu^2)^{2/3} e^{C\mu}. \tag{32}$$

The eventual threshold may depend on the chosen degree sequence. If in addition $\mu/\log A \to \infty$, the middle quotient diverges. With

$$\omega = e^{-2L^2/3969}, \qquad p_L = R^{-2}\omega M_L^{\mathrm{plat}}, \qquad X_L = (\omega/R)\max\{\mathcal{T}_L, 0\},$$

one nevertheless has

$$\lim_{L\to\infty} \frac{1}{L^2} \log \frac{X_L}{p_L^{2/3}} = -\frac{2}{11907}. \tag{33}$$

Proof. The argument retains both opposed caps in the numerator and the full plateau in the denominator.

Exact cross sections. For $G(s,z) = F(R^2 s, aRz)$ define

$$W_a(z) = -2\pi a z \vartheta(a(|z|-1)),$$

$$\mathcal{A}_a(z) = \pi\left([(a+\delta_0)^2 - a^2z^2]_+ - [(a-\delta_0)^2 - a^2z^2]_+\right),$$

where $[r]_+ = \max(r,0)$, and set

$$\mathcal{S} = v\int_0^4 \chi(s) \int_{\mathbb{R}} W_a(z)|G(s,z)|^2\, dz\, ds, \qquad \mathcal{K} = \int_0^4 \int_{\mathbb{R}} \mathcal{A}_a(z)|G(s,z)|^3\, dz\, ds.$$

Integrating $\partial_2\phi_L$ on the transverse plane gives

$$\int_{\mathbb{R}^2} \partial_2\phi_L(x_1, y, x_3)\, dx_1 dx_3 = -2\pi y \vartheta(|y|/R - a).$$

Indeed polar integration of the radial derivative reduces this integral to $2\pi y \int_{|y|}^{\infty} \partial_r \vartheta(r/R - a)\, dr$. The term involving $F\partial_3\phi_L$ in (29) integrates to zero, and so does the $B^3\partial_2\phi_L$ term. The remaining changes of variables $t = R^2 s$, $y = aRz$, together with $B = av/R$, give the exact relations

$$\mathcal{T}_L = \frac{a^2R^3}{2}\mathcal{S}, \qquad M_L^{\mathrm{plat}} = aR^5\mathcal{K}. \tag{34}$$

The second relation uses the full difference of disks represented by $\mathcal{A}_a$, including its outer caps.

Paired cap sign. Let $\gamma_L = \beta a/A$, so $\gamma_L \to \beta$. At $z = \pm(1 + r/a)$ the evenness of P_s in Proposition 4.3 gives the ideal values

$$2Q_{m,A}(s, r + \delta_0 + \gamma_L s), \qquad 2Q_{m,A}(s, r + \delta_0 - \gamma_L s),$$

respectively. Theorem 4.1, uniformly for $3 \le s \le 4$ and r in the fixed support of ϑ, implies

$$\frac{1}{\mu} \log \frac{Q_{m,A}(s, r + \delta_0 + \gamma_L s)}{Q_{m,A}(s, r + \delta_0 - \gamma_L s)} = 2G_s\gamma_L s + o(1).$$

The right side has a fixed positive lower bound. Both ideal values are positive and grow at the $\exp(F_s\mu^2)$ scale, so the absolute error in (26) is negligible relative to them. Hence the squared exact value at the negative cap is at most $e^{-c_1\beta\mu}$ times the squared value at the positive cap, for some $c_1 > 0$ and all sufficiently large parameters. Since W_a is odd, $W_a(z) \le 0$ for $z \ge 0$, and $v < 0$, the sum of each paired contribution is nonnegative throughout $3 \le s \le 4$.

The terminal contribution and earlier times. Let $h_\mu = (1+\mu^2)^{-1}$. On

$$4 - h_\mu \le s \le 4, \qquad r_c - h \le r \le r_c + h,$$

the time cutoff is one for sufficiently large parameters and $-W_a(1 + r/a) \ge c_2 a$. The positive ideal coordinate is at least $c_- + 4\gamma_L - Ch_\mu$. The derivative estimates in Lemma 4.2 give

$$|G(s, 1 + r/a)| \ge c_3 Q_{m,A}(4, c_- + 4\gamma_L).$$

The loss from its paired cap tends to zero. Using $dz = dr/a$ therefore gives a contribution at least $c_4(\beta/A)h_\mu Q_{m,A}(4, c_- + 4\gamma_L)^2$.

All real arguments arising from the fixed collar on $0 \le s \le 3$ have absolute value at most $1 + C_1/A$ for a fixed $C_1 > 0$. The global majorant (22) and the transfer bound thus bound the absolute contribution of this whole earlier interval by $C_2(\beta/A)Q_{m,A}(3, C_1)^2$. Theorem 4.1 gives

$$\log \frac{Q_{m,A}(4, c_- + 4\gamma_L)}{Q_{m,A}(3, C_1)} = (F_4 - F_3)\mu^2 + o(\mu^2), \qquad F_4 > F_3.$$

Thus the ratio of the early absolute bound to the terminal lower bound is at most

$$Ch_\mu^{-1} \exp(-2(F_4 - F_3)\mu^2 + o(\mu^2)),$$

which tends to zero. The same global majorant on the full interval gives an upper bound. Consequently, for one fixed $C_+ > 0$,

$$0 < c\frac{\beta}{A} h_\mu Q_{m,A}(4, c_- + 4\gamma_L)^2 \le \mathcal{S} \le C\frac{\beta}{A} Q_{m,A}(4, C_+)^2. \tag{35}$$

The constants here use $\int |W_a(z)|\, dz \le C$, as follows by setting $z = \pm(1 + r/a)$ on its fixed-width radial support.

Upper and lower payment on the full plateau. In the coordinate $z = e_a x$ the plateau projects to $|x| \le 1 + 2\delta_0/A$. Transport changes the ideal real coordinate by at most $4\gamma_L/A$. Therefore (22) and the transfer estimate give

$$\sup_{0 \le s \le 4,\ \mathcal{A}_a(z) > 0} |G(s, z)| \le CQ_{m,A}(4, c_p + 4\gamma_L).$$

Since $\mathcal{A}_a \le 4\pi a \delta_0$ and its support has bounded length in z,

$$\mathcal{K} \le CaQ_{m,A}(4, c_p + 4\gamma_L)^3. \tag{36}$$

For a lower bound, restrict only the integration to the outer strip

$$z = 1 + (c - \delta_0)/a, \qquad \delta_0 \le c \le \tfrac{3}{2}\delta_0, \qquad 4 - h_\mu \le s \le 4.$$

Here $\mathcal{A}_a(z) \ge c_5 a$ and $dz = dc/a$. The positive real field, the transfer estimate, and Lemma 4.2 yield

$$\mathcal{K} \ge ch_\mu Q_{m,A}(4, \delta_0 + 4\gamma_L)^3. \tag{37}$$

Only this lower bound restricts to a strip; the denominator itself in (30) remains the full plateau integral.

By (19), uniformly for the varying but bounded arguments here,

$$\log \frac{Q_{m,A}(4, c_- + 4\gamma_L)}{Q_{m,A}(4, c_p + 4\gamma_L)} = (2^{-4/3}\Delta_c + o(1))\mu, \qquad \log \frac{Q_{m,A}(4, C_+)}{Q_{m,A}(4, \delta_0 + 4\gamma_L)} \le C\mu.$$

Now (34) gives

$$R^{1/3} \frac{\mathcal{T}_L}{(M_L^{\mathrm{plat}})^{2/3}} = \frac{a^{4/3}}{2} \frac{\mathcal{S}}{\mathcal{K}^{2/3}}.$$

Combining the lower bound for $\mathcal{S}$ with (36) proves (31); combining the upper bound with (37) proves (32). Their polynomial factors have logarithms $O(\log A)$ because $\mu = o(A)$, so the stated divergence under $\mu/\log A \to \infty$ follows.

Finally the exact normalization is

$$\frac{X_L}{p_L^{2/3}} = \omega^{1/3} R^{1/3} \frac{\mathcal{T}_L}{(M_L^{\mathrm{plat}})^{2/3}}.$$

Both logarithmic bounds for the second factor are $O(\mu + \log A) = o(L^2)$, whereas $L^{-2} \log \omega^{1/3} = -2/11907$. This proves (33). □

Remark 4.6 (Scope of the exponent)**.** At a fixed slice the Chebyshev edge exponent is of size $m/\sqrt{A} = \mu^{3/2}$. Theorem 4.5 gives a residual exponential factor of size $\exp(\Theta(\mu))$ after time integration and division by the full plateau payment, with the explicit polynomial factors displayed in the theorem. Describing the logarithm of the entire quotient as $\Theta(\mu)$ additionally requires $\log A = o(\mu)$. The negative normalized rate applies to this one family, with its specified coefficients, drift, and degree range. It is not a uniform estimate over all packets and gives no regularity or singularity conclusion for general Navier–Stokes solutions.

Remark 4.7 (Degrees outside the proof)**.** When m is comparable to or larger than A^2, the relation $\mu/A \to 0$ fails and the edge saddle used above is no longer the relevant expansion. In addition, the error estimate in Proposition 4.3 contains $\exp(Cm + Cm^2/A^2)$. It does not tend to zero from the argument given here in such a range. In particular a formal calculation at $m \asymp A^4$ is not a consequence of any theorem in this paper.

5 Observation frame and the velocity in the denominator

The distinction between the entire spatial plateau and the complete velocity is essential. In this section write $M_{F,L} = M_L^{\mathrm{plat}}$ for (30), and retain the exact initial coefficients, cutoffs, and degree sequences of Theorem 4.5 unless stated otherwise. The following consequences concern the normalization of that explicit family. They are not general local-energy estimates.

Proposition 5.1 (A change of observation frame)**.** *For $u(t,x) = (0, B, F(t,x_2))$, define $y = x - Bte_2$ and*

$$\widetilde{u}(t,y) = u(t, y + Bte_2) - Be_2 = (0, 0, H(t,y_2)).$$

Then $H_t = H_{y_2y_2}$. For any smooth radial cutoff $\phi(y)$ supported in a Euclidean coordinate chart of the torus,

$$\frac{1}{2}\int_{\mathbb{T}^3} |\widetilde{u}|^2 \widetilde{u} \cdot \nabla\phi \, dy = 0.$$

Proof. The definition gives $H(t,y_2) = F(t, y_2 + Bt)$, so $H_t = F_t + BF_{x_2} = F_{x_2x_2}$. The flux integrand is $H(t,y_2)^3 \partial_{y_3}\phi/2$. Its integral in y_3 vanishes by the fundamental theorem of calculus and the support of ϕ. Periodicity gives the same conclusion. □

A cutoff fixed in y corresponds to a translating cutoff in the original coordinates. Moreover the kinetic energy changes when the constant velocity is subtracted. Thus this zero flux and the positive flux in Theorem 4.5 are different observations; the latter is not Galilean invariant. It measures constant transport through a fixed window. The construction does not introduce a vortex-stretching or energy-cascade mechanism.

Proposition 5.2 (Complete velocity mass for the stated amplitude)**.** *Define the positive full-velocity mass and the spacetime volume by*

$$M_{u,L} = \int_0^{4R^2} \int_{\mathcal{P}_L} (B^2 + F(t,x_2)^2)^{3/2} \, dx \, dt, \qquad V_L = 4R^2 |\mathcal{P}_L|.$$

Along every degree sequence (17),

$$V_L = 32\pi\delta_0 a^2 R^5 \left(1 + \frac{\delta_0^2}{3a^2}\right), \tag{38}$$

$$M_{u,L} = (1 + o(1))\, 32\pi\delta_0\beta^3 \frac{a^5}{A^3} R^2, \tag{39}$$

$$M_{F,L} = R^5 \exp(o(L^2)), \qquad \mathcal{T}_L = R^3 \exp(o(L^2)). \tag{40}$$

The final expression uses the eventual positivity of $\mathcal{T}_L$. In particular

$$R^{1/3}\frac{\mathcal{T}_L}{M_{u,L}^{2/3}} = R^2\exp(o(L^2)) \longrightarrow 0, \qquad \lim_{L\to\infty}\frac{1}{L^2}\log\left(R^{1/3}\frac{\mathcal{T}_L}{M_{u,L}^{2/3}}\right) = -\frac{9}{20000}. \tag{41}$$

All limits refer to the amplitudes fixed in (24); they are not uniform over arbitrary rescalings of those amplitudes.

Proof. Subtracting the volumes of the two balls gives (38) exactly. Since $a, A \asymp L$ and $\mu = o(A)$, the global majorant and dynamic transfer give

$$\sup_{[0,4R^2]\times\mathcal{P}_L}|F| \le CQ_{m,A}(4, c_p + 4\gamma_L) + o(1) = \exp(o(L^2)), \qquad \gamma_L = \beta a/A.$$

Indeed the heat-edge theorem gives, for every bounded coordinate c, $\log Q_{m,A}(4,c) = F_4\mu^2 + o(\mu^2) + O(\mu)$ uniformly in c. On the other hand $|B| = \beta a/(AR)$, and hence $\sup|F|/|B| \to 0$. Uniformly on the entire integration region, $(B^2+F^2)^{3/2} = |B|^3(1+o(1))$. Multiplication by the exact volume proves (39).

For the remaining two scales, use the positive lower and upper bounds (35), (36), and (37), followed by (34). Every $Q(4,c)$ in those bounds has logarithm $o(L^2)$; a, A, β, and $h_\mu = (1+\mu^2)^{-1}$ contribute logarithms $O(\log A) = o(L^2)$. Both sides therefore give (40). This step uses a lower bound on the whole signed time integral, not an absolute value of a cap contribution.

The denominator $M_{F,L}$ is positive for every nonzero member of the family: a nonzero finite Fourier sum cannot vanish on a spatial interval, and the plateau has positive cross-sectional area on an interval. The time evolution preserves its nonzero Fourier coefficients. The positivity of $\mathcal{T}_L$ is only eventual and comes from paired late-time caps and absorption of the earlier signed contribution in Theorem 4.5. Thus the logarithm in (41) is defined eventually along each sequence. Equations (39)–(40) give the net power $R^{1/3+3-4/3} = R^2$. Finally $2L^{-2}\log R = -9/20000$. □

The qualifier about amplitude cannot be removed. The next elementary observation makes that dependence explicit.

Proposition 5.3 (Amplitude rescaling)**.** *Fix one shear, a finite observation region of spacetime volume $V > 0$, and the same numerator and denominator conventions as above. Assume $M_F > 0$ and $\mathcal{T} \ge 0$. For $\lambda > 0$ replace its initial profile by $\lambda F(0,\cdot)$ and keep B fixed. The resulting solution has profile $\lambda F(t,\cdot)$ throughout the same interval. Set*

$$Z_F = R^{1/3}\frac{\mathcal{T}}{M_F^{2/3}}, \qquad Z_u(\lambda) = R^{1/3}\frac{\lambda^2\mathcal{T}}{\left[\int(B^2+\lambda^2F^2)^{3/2}\right]^{2/3}}.$$

Then $Z_u(\lambda)$ is nondecreasing and

$$\frac{Z_F}{1+(B/\lambda)^2(V/M_F)^{2/3}} \le Z_u(\lambda) \le Z_F, \qquad \lim_{\lambda\to\infty} Z_u(\lambda) = Z_F. \tag{42}$$

For the family of Proposition 5.2, the explicit choice $\lambda_L = R^{-2}$ gives $Z_u(\lambda_L)/Z_F \to 1$. Consequently that rescaled family has $Z_u(\lambda_L) \to \infty$ if $\mu/\log A \to \infty$.

Proof. The scalar advection–heat equation is linear, and its transport flux is quadratic in F. After dividing the full mass by λ^3, its two-thirds power is

$$\left\|F^2 + (B/\lambda)^2\right\|_{L^{3/2}}.$$

It decreases with λ and is bounded below by $\|F^2\|_{L^{3/2}} = M_F^{2/3}$. Minkowski's inequality bounds it above by $M_F^{2/3} + (B/\lambda)^2 V^{2/3}$. This proves the inequalities, monotonicity, and limit, including $B = 0$, when equality holds throughout. Dominated convergence also gives the limit directly, with dominator $(F^2 + B^2)^{3/2}$ for $\lambda \geq 1$. In the specified family,

$$(B/\lambda_L)^2 (V_L/M_{F,L})^{2/3} = R^2 \exp(o(L^2)) \longrightarrow 0.$$

The asserted divergence now follows from Theorem 4.5. Each rescaled member still uses one initial datum over the entire time interval. □

This rescaling retains the very large constant drift and does not produce uniformly bounded initial energy. It shows why (41) cannot be read as an impossibility theorem for every amplitude normalization. It is also not a counterexample to a general local-energy estimate: the component estimate in this paper retains its stated geometric and frequency factors, and rescaling does not remove them.

Proposition 5.4 (Bounded drift for the same initial shape)**.** *Keep a, A, R, m and the initial coefficients (24), but replace the drift by any B_L with $|B_L| \leq B_* < \infty$. Denote the corresponding flux and component mass by $\mathcal{T}_L^{(B_L)}$ and $M_{F,L}^{(B_L)}$. There is a constant C, depending on the fixed cutoffs and B_*, such that eventually along every allowed degree sequence*

$$R^{1/3} \frac{|\mathcal{T}_L^{(B_L)}|}{(M_{F,L}^{(B_L)})^{2/3}} \leq C|B_L| R a^{1/3} (1+\mu^2)^{2/3} e^{C\mu} \longrightarrow 0. \tag{43}$$

The same upper bound holds after arbitrary positive amplitude rescaling and replacement of the component mass by the complete velocity mass. If $B_L = 0$, the flux is exactly zero.

Proof. In Proposition 4.3 the real shift becomes $B_L Rs/A$ in the x coordinate. All other operator, carrier, and confluence estimates are unchanged; the shift is smaller than the one already permitted there. Thus the same absolute transfer bound holds uniformly for bounded B_L.

The exact flux cross section and $\int |W_a| \leq C$ give

$$|\mathcal{T}_L^{(B_L)}| \leq C|B_L| a R^4 Q_{m,A}(4, C_1)^2$$

for a fixed $C_1 > 0$. For the mass lower bound choose the outer plateau strip on the side whose displacement is $+|B_L|Rs$ in the edge coordinate. Parity of the ideal polynomial permits either side. On its terminal time layer of length h_μ, its edge coordinate is at least δ_0. The terminal derivative estimates and the absolute transfer bound consequently give

$$M_{F,L}^{(B_L)} \geq c a R^5 h_\mu Q_{m,A}(4, \delta_0)^3.$$

This lower bound restricts the integration only to prove the inequality; the defined mass still includes the full plateau and time interval. The heat-edge ratio between C_1 and δ_0 is at most $e^{C\mu}$. Combining the last two bounds gives (43). Its logarithmic upper bound is $\log R + O(\mu + \log A)$ plus a bounded term, which tends to $-\infty$ because $\mu = o(A)$ and $A \asymp L$. The component quotient is unchanged by amplitude rescaling, whereas the full mass is at least the component mass. These facts give the last assertion. At $B_L = 0$ the exact cross-sectional flux formula vanishes. □

This last proposition concerns the same initial shape and scale. It neither rules out other profiles with bounded drift nor supplies matching dynamic upper and lower bounds over the general Fourier class.

6 A uniform bound for the complete collar flux

This auxiliary application combines established spatial and temporal inequalities with the shear energy identity. It is not a separate claim of a new extrapolation scale.

Lemma 6.1 (Spatial input from Zhang)**.** *For $g(x) = \sum_{j=1}^{N} c_j e^{i\xi_j x}$ with arbitrary real frequencies, complex coefficients, and $0 \le d \le 1$,*

$$\max\{|g(1+d)|, |g(-1-d)|\} \le C_Z N e^{3\sqrt{2}N\sqrt{d}} \|g\|_{L^2(-1,1)}, \qquad C_Z = \sqrt{9 \cdot 8191/2}. \tag{44}$$

Consequently, for $J = [-e, e]$ and $I = [-e(1+d), e(1+d)]$, $e > 0$,

$$\|g\|^2_{L^\infty(I)} \le \frac{C_Z^2}{e} N^2 e^{6\sqrt{2}N\sqrt{d}} \int_J |g|^2. \tag{45}$$

Attribution and rescaling. The right endpoint assertion is Zhang's near-edge theorem [10, Theorem 1.1 and Proposition 4.2, v1], using the admissible finite-range constant 8191. Reflection proves the left endpoint assertion. There is no frequency separation assumption; repeated frequencies are merged and zero coefficients are allowed. Affine rescaling proves the second assertion outside J. At an interior point z, applying the $d = 0$ endpoint bound separately to $[-e, z]$ and $[z, e]$ gives $\ell_\pm |g(z)|^2 \le 2C_Z^2 N^2 \int_{J_\pm} |g|^2$. Adding these estimates gives the same bound on J. □

The theorem cited here is a preprint result and is used as an attributed input. The required range in this section is only $d \le 1$. Corollary A.1 supplies the temporal input and (65) its rescaled form.

6.1 The physical model and the uniform theorem

Write $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$. Fix $0 < \delta_0 < \delta$ and a function $\vartheta \in C_c^\infty((-\delta, \delta))$ such that $0 \le \vartheta \le 1$ and $\vartheta = 1$ on $[-\delta_0, \delta_0]$. Fix also $\eta \in C^1([0,4])$ with $0 \le \eta \le 1$ and $\eta(0) = 0$. For $R > 0$, $a \ge \max(1, \delta + 2\delta_0)$ and $(a+\delta)R < \pi/2$, set

$$\xi_{a,R}(x) = \vartheta(|x|/R - a), \qquad \mathcal{S}^{\mathrm{plat}}_{a,R} = \{x \in \mathbb{T}^3 : \, ||x|/R - a| \le \delta_0\}, \tag{46}$$

$$F(t, x_2) = \sum_{j=1}^{q} A_j e^{-n_j^2 t} \cos(n_j x_2 - \phi_j - n_j B t), \tag{47}$$

where $q \ge 1$, $1 \le n_1 < \cdots < n_q \le 2n_1$ are integers, $A_j \ge 0$, and $\phi_j, B \in \mathbb{R}$. The support of the cutoff lies in a Euclidean chart of the torus, and it is extended by zero outside that chart. The field $u = (0, B, F)$ is an exact smooth Navier–Stokes solution with viscosity one and pressure zero: its only nonzero evolution equation is $F_t + BF_{x_2} - F_{x_2x_2} = 0$.

Define the complete flux and the full plateau observation by

$$\mathcal{T}_{\boldsymbol{n},R} = \frac{1}{2} \int_0^{4R^2} \int_{\mathbb{T}^3} \eta(t/R^2) B \, \partial_2 \xi_{a,R}(x) F(t, x_2)^2 \, dx \, dt, \tag{48}$$

$$M^{\mathrm{plat}}_{\boldsymbol{n},R} = \int_0^{4R^2} \int_{\mathcal{S}^{\mathrm{plat}}_{a,R}} |F(t, x_2)|^3 \, dx \, dt. \tag{49}$$

The time cutoff occurs in the flux; the plateau observation uses the entire unweighted clock.

Theorem 6.2 (Complete-clock upper bound)**.** *For the data above, put*

$$\Delta_a = \frac{\delta + \delta_0}{a - \delta_0}, \qquad \Phi_a = 12\sqrt{2} q \sqrt{\Delta_a}.$$

There is a constant C, depending only on $\vartheta, \delta_0, \delta$ and $\|\eta'\|_\infty$, such that

$$|\mathcal{T}_{\boldsymbol{n},R}| \le Ca^{2/3}R^{-1/3}\left(q^4 + \frac{q^7}{a^2} + q^{13/3}\right) e^{\Phi_a}\big(M^{\mathrm{plat}}_{\boldsymbol{n},R}\big)^{2/3}. \tag{50}$$

In particular,

$$|\mathcal{T}_{\boldsymbol{n},R}| \le Ca^{2/3}R^{-1/3}q^7 e^{\Phi_a}\big(M^{\mathrm{plat}}_{\boldsymbol{n},R}\big)^{2/3}. \tag{51}$$

The constant is independent of q, R, all frequencies, amplitudes, phases, and B.

Proof. Use the scaled variables

$$s = t/R^2, \quad z = x_2/(aR), \quad G(s,z) = F(R^2 s, aRz),$$
$$\kappa_j = n_j aR, \quad \alpha = \kappa_1, \quad v = BR/a, \quad \lambda = (\alpha/a)^2.$$

Then $G_s + vG_z - a^{-2}G_{zz} = 0$, and its $2q$ complex spatial frequencies are $\pm\kappa_j$, with $\alpha \le \kappa_j \le 2\alpha$. Let

$$e_a = 1 - \delta_0/a, \quad E_a = [-e_a, e_a], \quad I_a = [-1 - \delta/a, 1 + \delta/a],$$
$$h(s) = \int_{E_a} |G(s,z)|^3\, dz, \quad H = \int_0^4 h(s)\, ds.$$

The assumptions imply $e_a \ge 1/2$, $\Delta_a \le 1$, and uniformly bounded upper and lower lengths for these intervals. Equation (45) with $N = 2q$, followed by Hölder, gives

$$\|G(s)\|^2_{L^\infty(I_a)} \le \frac{4C_Z^2}{e_a} q^2 e^{\Phi_a} \int_{E_a} |G|^2 \le Cq^2 e^{\Phi_a} h(s)^{2/3}. \tag{52}$$

The following two estimates from the literature are used in the remaining argument. Erdélyi's Markov inequality [2, Theorem 9.1, arXiv v1] states, for $f(x) = \sum_{r=1}^N c_r e^{i\mu_r x}$ with $\mu_r \in \mathbb{R}$ on $[0,1]$, that

$$\|f'\|_\infty \le C\left(N^5 + \sum_{r=1}^N \mu_r^2\right)^{1/2} \|f\|_\infty. \tag{53}$$

The more precise expression in that paper is $(1+\varepsilon_N)(108N^5 + \sum_r \mu_r^2)^{1/2}$, with a uniformly bounded factor $1 + \varepsilon_N$. Kós's endpoint inequality [6], recorded in Erdélyi's paper as (1.2), is

$$|f(0)| \le 2N\|f\|_{L^2(0,1)}, \qquad f(r) = \sum_{j=1}^N c_j e^{\gamma_j r}, \quad \operatorname{Re}\gamma_j \ge 0. \tag{54}$$

Map I_a affinely onto $[0,1]$ in (53). Its length is bounded above and below, and $\sum_j(\kappa_j^2 + (-\kappa_j)^2) \le 8q\alpha^2$. Together with (52), this proves

$$\|G_z(s)\|^2_{L^\infty(I_a)} \le C(q^7 + q^3\alpha^2) e^{\Phi_a} h(s)^{2/3}. \tag{55}$$

For fixed z, the function $r \mapsto G(4-r, z)$, $0 \le r \le 1$, has at most $2q$ exponential terms with exponents $(\kappa_j/a)^2 \pm iv\kappa_j$. Applying (54), cubing, and using Hölder on the unit interval yields

$$h(4) \le 64q^3 \int_3^4 h(s)\, ds \le 64q^3 H, \qquad h(4)^{2/3} \le 16q^2 H^{2/3}. \tag{56}$$

The radial geometry can be computed exactly. At $y = x_2$, the cross-sectional integral of the cutoff is $2\pi \int_{|y|}^\infty r\vartheta(r/R - a)\, dr$. Differentiating with respect to y gives

$$\int_{\mathbb{T}^2} \partial_2 \xi_{a,R}(x_1, y, x_3)\, dx_1\, dx_3 = -2\pi y \vartheta(|y|/R - a).$$

Thus, with

$$W_a(z) = -2\pi a z \vartheta(a(|z|-1)), \qquad \Xi_a(z) = \int_{-\infty}^{z} W_a(r)\,dr,$$

the exact flux conversion is

$$\mathcal{T}_{\boldsymbol{n},R} = \frac{a^2R^3}{2}\, v \int_0^4 \eta(s) \int_{\mathbb{R}} W_a(z) G(s,z)^2\,dz\,ds. \tag{57}$$

The function W_a is odd and has zero integral. Its two support intervals have length $O(a^{-1})$, and $|W_a| \le Ca$ there. Consequently

$$\operatorname{supp}\Xi_a \subset I_a, \qquad \|\Xi_a\|_1 + \|\Xi_a\|_\infty \le C, \qquad \|\Xi_a''\|_1 = \|W_a'\|_1 \le Ca. \tag{58}$$

These bounds follow directly by differentiating W_a on its support; that support stays away from $z = 0$.

Put $\mathcal{E}(s) = \int_{\mathbb{R}} \Xi_a(z) G(s,z)^2\,dz$. Multiplying $(G^2)_s + v(G^2)_z - a^{-2}(G^2)_{zz} = -2a^{-2}|G_z|^2$ by Ξ_a, and then integrating in space and time, gives

$$\begin{aligned} v\int_0^4 \eta \int W_a G^2 = \eta(4)\mathcal{E}(4) - \int_0^4 \eta'\mathcal{E}\,ds \\ - a^{-2}\int_0^4 \eta \int \Xi_a'' G^2 + 2a^{-2}\int_0^4 \eta \int \Xi_a |G_z|^2. \end{aligned} \tag{59}$$

There is no initial-face term because $\eta(0) = 0$. Using (52), (56), and (58), the first three terms on the right have absolute sum at most

$$C(q^4 + q^2 + a^{-1}q^2)e^{\Phi_a} H^{2/3}. \tag{60}$$

Here $\int_0^4 h^{2/3} \le 4^{1/3} H^{2/3}$. The last term, by (55), is bounded by

$$Ce^{\Phi_a}\left[\frac{q^7}{a^2}\int_0^4 \eta h^{2/3}\,ds + q^3\lambda \int_0^4 \eta h^{2/3}\,ds\right].$$

The first summand is at most $Cq^7a^{-2}H^{2/3}$. If $\lambda \le 1$, Hölder bounds the second by $Cq^3H^{2/3}$. If $\lambda > 1$, the temporal exponents of G have real parts in $[\lambda, 4\lambda]$ and $\eta(s) \le \|\eta'\|_\infty s$. Equation (65), with $N \le 2q$, bounds that summand by $Cq^{13/3}\lambda^{-1/3}H^{2/3} \le Cq^{13/3}H^{2/3}$. Combining these estimates with (59) gives

$$\left| v\int_0^4 \eta \int W_a G^2 \right| \le C\left(q^4 + \frac{q^7}{a^2} + q^{13/3}\right) e^{\Phi_a} H^{2/3}. \tag{61}$$

Finally, the plateau cross-sectional area at $x_2 = aRz$, divided by R^2, equals

$$\pi\left([(a+\delta_0)^2 - a^2z^2]_+ - [(a-\delta_0)^2 - a^2z^2]_+\right).$$

For $z \in E_a$ it is exactly $4\pi a\delta_0$. As $dt\,dx_2 = aR^3\,ds\,dz$, this proves

$$M^{\mathrm{plat}}_{\boldsymbol{n},R} \ge 4\pi\delta_0 a^2 R^5 H. \tag{62}$$

Equations (57), (61), and (62) prove (50). Since $a, q \ge 1$, $q^4 + q^7/a^2 + q^{13/3} \le 3q^7$, proving (51). □

Remark 6.3 (A shrinking-gap regime)**.** Let $a = pL$ with fixed $p > 0$, and let $\omega = \exp(-\gamma L^2)$ with fixed $\gamma > 0$. Set $P = R^{-2}\omega M^{\mathrm{plat}}_{\boldsymbol{n},R}$ and $X = (\omega/R)[\mathcal{T}_{\boldsymbol{n},R}]_+$. Equation (51) becomes

$$X \le Ca^{2/3}q^7\omega^{1/3}e^{12\sqrt{2}q\sqrt{\Delta_a}}P^{2/3}.$$

If $q = q(L) = o(L^{5/2})$, then $\Delta_a = O(L^{-1})$ and

$$\lim_{L\to\infty}\frac{1}{L^2}\log\left(Ca^{2/3}q(L)^7\omega^{1/3}e^{12\sqrt{2}q(L)\sqrt{\Delta_a}}\right) = -\frac{\gamma}{3}.$$

This is the rate of the proved upper-bound coefficient. It does not assert that a particular flux attains that rate. For the normalization $\gamma = 2/3969$, the rate is $-2/11907$.

7 Limitations and computational assistance

The quantitative edge criterion requires a positive exterior profile, a global envelope on both sides, an interior bound, and a lower bound in a fixed saddle neighborhood. The normalized first and second Chebyshev families meet these hypotheses. The resulting error orders are sufficient, not known to be best. The criterion does not establish an arbitrary-parameter Jacobi theorem or cover degrees with μ/A bounded away from zero.

The fixed-data shear lower bound and the uniform dyadic upper bound have different quantifiers. The lower bound does not match every polynomial or exponential factor of the upper coefficient. Matching dynamic upper and lower bounds for the same observable and class remain open. The sharp $N^{4/3}$ order of the abstract temporal inequality does not settle this spatially constrained question. Replacing a signed flux by the integral of its pointwise absolute value would also change the observable and its cancellation argument.

The original-amplitude complete-velocity quotient, its restoration by large amplitude rescaling, and the bounded-drift estimate are all statements about specified shear families. The drift and the torus energy need not be bounded along the main sequence. The planar extension generally has infinite total energy. No uniform energy class, limiting weak solution, vortex-stretching mechanism, or criterion for arbitrary three-dimensional Navier–Stokes data is established. The constant $-2/11907$ in the optional normalization is the exponent of a chosen external weight, not a universal dynamical constant.

The special-function literature comparison checks concrete primary sources and parameter regimes. It is not an exhaustive priority search; more general terminating hypergeometric results may exist. No priority claim is made beyond the explicit comparisons in Section 3.

Computer and AI assistance. OpenAI Codex assisted with drafting, literature retrieval, proof organization, exact finite algebra checks, and verification scripts. The theorem proofs are written in this paper, and imported inequalities are stated with their hypotheses and attributed. Separate automated review runs checked the new quantitative theorem, normalization arguments, and temporal order proof; these are not human peer review or formal proof-assistant certification. The author remains responsible for the manuscript. Finite computations test identities and selected parameters, not the continuum asymptotic assertions.

Data and code availability. The ancillary files contain exact finite-algebra checks and the code, parameters, data, and reference SVG for Figure 1. The file `anc/README.md` gives the commands and dependencies. These reproducibility materials complement the proofs; finite checks and quadrature diagnostics do not certify continuum limits.

A A time-weighted literature corollary and its order

Corollary A.1 (Time-weighted decay from Erdélyi)**.** *Let (E,ν) be a measure space, $N \ge 1$, and $\zeta_j \in \mathbb{C}$ with $\operatorname{Re}\zeta_j \ge 1$. For measurable finite coefficients define*

$$f(t,z)=\sum_{j=1}^{N} c_j(z)e^{-\zeta_j t}, \quad k(t)=\int_E |f(t,z)|^3\,d\nu(z), \quad K_T=\int_0^T k(t)\,dt.$$

If $T>0$ and $K_T<\infty$, then

$$\int_0^T t\,k(t)^{2/3}\,dt \le C_* N^{4/3} K_T^{2/3}, \qquad C_*=4^{-1/3}36^{4/3}+1. \tag{63}$$

No upper bound on the real parts or bound on the imaginary parts is needed.

Proof. Merge coincident exponents; estimates with the original N follow by monotonicity. Erdélyi's infinite–finite range theorem [2, Theorem 3.1, arXiv v1] applies to complex exponential sums with nonpositive real parts. Applying it to $e^t f(t,z)$ gives

$$J(z) := \int_0^\infty e^t |f(t,z)|^2\,dt \le 8191 \int_0^{9N} e^t |f(t,z)|^2\,dt \le 8191 e^{9N} I_{9N}(z),$$

where $I_b(z) = \int_0^b |f(t,z)|^2\,dt$. These integrals are finite pointwise because the coefficients are finite and the real parts of the decay rates are at least one. The midpoint estimate [2, Lemma 12.5, arXiv v1], on an interval of length two, applied to $e^{t/2} f(t,z)$, gives for $t \ge 1$

$$e^t |f(t,z)|^2 \le N \int_{t-1}^{t+1} e^u |f(u,z)|^2\,du \le N J(z).$$

Thus $|f(t,z)|^2 \le 8191 N e^{9N-t} I_{9N}(z)$. This midpoint estimate allows arbitrary complex exponents.

Take $S = 36N$. If $T \ge S$, Minkowski and Hölder give

$$\|I_{9N}\|_{L^{3/2}(E)} \le \int_0^{9N} k(t)^{2/3}\,dt \le (9N)^{1/3} K_T^{2/3}.$$

The integral over $[S,T]$ is therefore at most

$$8191 N (9N)^{1/3} (36N+1) e^{-27N} K_T^{2/3} \le N^{4/3} K_T^{2/3}.$$

For the last inequality, $(36N+1)e^{-27N}$ decreases for $N \ge 1$, and $8191\, 9^{1/3} 37 e^{-27} < 1$. For either $T \ge S$ or $T < S$, the initial part is bounded by

$$\int_0^{\min(T,S)} t k(t)^{2/3}\,dt \le \left(\int_0^S t^3\,dt \right)^{1/3} K_T^{2/3} = 4^{-1/3} (36N)^{4/3} K_T^{2/3}.$$

This proves the claim, including $K_T = 0$. □

Proposition A.2 (An explicit order witness)**.** *For each $N \ge 1$ there is a real sum with exactly N nonzero terms, all decay rates in $[1, 5/4]$, such that, on the single-point space and with $T = N$,*

$$\frac{\int_0^N t f_N(t)^2\,dt}{\left(\int_0^N f_N(t)^3\,dt \right)^{2/3}} \ge \frac{N^{4/3}}{128}. \tag{64}$$

Consequently the order $N^{4/3}$ in (63) cannot be replaced by $o(N^{4/3})$ in its abstract class.

Proof. Set $\epsilon = N^{-2}$, $w(t) = (1 - e^{-\epsilon t})/\epsilon$, and

$$f_N(t) = e^{-t} \sum_{k=0}^{N-1} \frac{w(t)^k}{k!} = \sum_{j=0}^{N-1} b_j e^{-(1+j\epsilon)t}, \qquad b_j = \frac{(-1)^j}{j! \epsilon^j} \sum_{h=0}^{N-1-j} \frac{1}{\epsilon^h h!}.$$

The coefficient identity follows by the binomial theorem and all its inner summands are positive. Thus every $b_j \ne 0$, and the distinct decay rates lie in $[1, 5/4]$ since $(N-1)/N^2 \le 1/4$. Write $f_{0,N}(u) = e^{-u} \sum_{k=0}^{N-1} u^k/k!$. This is the probability that a Poisson variable of mean u is less than N, equivalently the elementary incomplete-gamma identity [7, §8.4.11]. For $0 \le t \le N/2$ one has

$$0 \le w(t) \le t, \qquad t - w(t) \le \epsilon t^2/2 \le 1/8, \qquad f_{0,N}(w(t)) \ge 1 - w(t)/N \ge \tfrac{1}{2}.$$

The middle inequality follows by integrating $1 - e^{-\epsilon u} \le \epsilon u$; the last one is Markov's probability inequality. Hence $f_N(t) = e^{-(t-w(t))} f_{0,N}(w(t)) \ge \frac{1}{2} e^{-1/8} > 1/4$ there. On all $t \ge 0$, $0 < f_N(t) \le f_{0,N}(t) \le 1$. Therefore $K_N = \int_0^N f_N^3 \, dt$ lies in $(0, N]$ and

$$\int_0^N t f_N^2 \, dt \ge \frac{1}{16} \int_0^{N/2} t \, dt = \frac{N^2}{128}.$$

Division proves (64). Nothing excludes $N = 1$: then $b_0 = 1$ and $f_1 = e^{-t}$. If unit cubic mass is desired, replace f_N by $K_N^{-1/3} f_N$; the quotient is unchanged. □

This elementary construction concerns the abstract temporal norm quotient. Its nearby real decay rates need not be the squared integer frequencies of a shear, and it has no corresponding spatial constraint. It does not prove optimality of the q^7 flux coefficient, a sharp exponential constant, or a dynamical transition. The upper estimate is a direct literature corollary; no priority claim is made for this order witness.

For later application, if the temporal rates of $G(s, z)$ have real parts at least $\lambda > 0$, write $h(s) = \int_E |G(s, z)|^3 \, d\nu$, $H = \int_0^4 h(s) \, ds$. For $0 \le \eta(s) \le C_\eta s$, substitution $t = \lambda s$ gives

$$\lambda \int_0^4 \eta(s) h(s)^{2/3} \, ds \le C_* C_\eta N^{4/3} \lambda^{-1/3} H^{2/3}. \tag{65}$$

B Classical spatial witnesses and elementary realization

This appendix records the supporting spatial facts with their provenance. The near-edge scale and Chebyshev confluence construction are prior work [10, Propositions 7.1 and 8.4]; polynomial endpoint witnesses also occur in Erdélyi [2]. The elementary modifications below impose real values and a dyadic band. They are not separate new extrapolation theorems. All norms here use Lebesgue measure on $[-1, 1]$.

Lemma B.1 (Coefficient identity)**.** *For a real polynomial p of degree n and $\eta > 0$,*

$$p\left(\frac{e^{i\eta x} - 1}{i\eta}\right) = \sum_{j=0}^{n} b_j(\eta) e^{ij\eta x}, \qquad b_j(\eta) = \frac{p^{(j)}(i/\eta)}{j!(i\eta)^j}. \tag{66}$$

If p is real-rooted, every b_j is nonzero for every $\eta > 0$. For a fixed real-rooted p, $M > 0$, and real θ, taking twice the real part after multiplication by $e^{i(Mx+\theta)}$ yields exactly $n + 1$ positive frequencies in $[M, 2M]$ when $n\eta \le M$, and tends locally uniformly to $2p(x)\cos(Mx + \theta)$ as $\eta \downarrow 0$.

Proof. Expand p about i/η, using $w_\eta(x) = i/\eta + e^{i\eta x}/(i\eta)$. Rolle's theorem makes the zeros of each nonconstant derivative real, so it cannot vanish at i/η. The top derivative is a nonzero constant. The positive frequencies after realification are $M + j\eta$, with amplitudes $2|b_j|$. Finally $|w_\eta(x) - x| \le \eta x^2/2$ proves the compact convergence. Allowing arbitrary phases means that positive amplitudes do not prevent cancellation: signs can be absorbed into the phases. □

Proposition B.2 (Known spatial orders retained in the real band)**.** *Let $\mathcal{R}_q^{\mathrm{dyad}}$ consist of real cosine sums with exactly $q \ge 2$ positive frequencies in one dyadic band, nonzero amplitudes, and arbitrary phases. For $0 < d \le 1$,*

$$\sup_{g \in \mathcal{R}_q^{\mathrm{dyad}}} \frac{|g(1 + d)|}{\|g\|_2} \ge \frac{e^{(q-1)\,\mathrm{arcosh}(1+d)}}{2\sqrt{2}},$$

$$\sup_{g \in \mathcal{R}_q^{\mathrm{dyad}}} \frac{\int_1^{1+d} |g|^2}{\|g\|_r^2} \ge \frac{d}{128} e^{2(q-1)\sqrt{7d/8}}, \qquad r = 2, 3,$$

$$\sup_{g\in\mathcal{R}_q^{\mathrm{dyad}}}\frac{|g(1)|}{\|g\|_2}\ge q/\sqrt{2},\qquad \sup_{g\in\mathcal{R}_q^{\mathrm{dyad}}}\frac{|g(1)|}{\|g\|_3}\ge 2^{-1/3}q^{2/3}.$$

These are supremum statements along confluent sequences.

Proof. For the exterior statements take $p=T_{q-1}$, $M=1$, $\theta=-(1+d)$ in Lemma B.1. The limit is $g_0=2T_{q-1}(x)\cos(x-1-d)$, with $\|g_0\|_2^2\le 8$ and $\|g_0\|_3^3\le 16$. At $1+d$ the carrier is one and $T_{q-1}(1+d)\ge\frac{1}{2}e^{(q-1)\operatorname{arcosh}(1+d)}$. On $[1+7d/8,1+d]$ the squared carrier is at least $1/2$ and T_{q-1} is increasing. The exterior squared integral is thus at least $(d/16)e^{2(q-1)\sqrt{7d/8}}$, using $\operatorname{arcosh}(1+x)\ge\sqrt{x}$ for $0\le x\le 1$. Compact convergence transfers both ratios.

For the endpoint take the classical Legendre kernel $K_{q-1}(x)=\sum_{r=0}^{q-1}(2r+1)P_r(x)/2$. Orthogonality gives $K_{q-1}(1)=\|K_{q-1}\|_2^2=q^2/2$; $|P_r|\le 1$ gives $\|K_{q-1}\|_3^3\le q^4/4$. This kernel has only simple roots in $(-1,1)$: it is orthogonal to all lower degree polynomials for the positive weight $1-x$, since $\int K_{q-1}r=r(1)$. Multiplying its sign-changing factors gives the usual contradiction if fewer than $q-1$ such roots occur. Apply the coefficient lemma with $M=1$, $\theta=-1$. The carrier is one at the observed endpoint and cannot increase either norm, which proves the last two inequalities. □

The squared L^3 endpoint lower power here is $q^{4/3}$. It does not prove the optimal endpoint power: Erdélyi's L^p estimate [2, Theorem 2.10, arXiv v1] gives a squared upper power $q^{5/3}$ at $p=3$, counting $2q$ complex branches.

Proposition B.3 (A prescribed-time integer realization)**.** *For a real-rooted p of degree $q-1$, $R,A>0$, $\eta=AR$, an integer $n_0\ge q-1\ge 1$, and fixed $s_*\ge 0,B,\theta$, put $n_j=n_0+j$, $a_j=2|b_j(\eta)|e^{n_j^2R^2s_*}$ and $\phi_j=-\theta-\arg b_j(\eta)-n_jBR^2s_*$. Then the exact shear $F(t,y)=\sum_j a_je^{-n_j^2t}\cos(n_jy-\phi_j-n_jBt)$ satisfies*

$$F(R^2s_*,ARx)=2\operatorname{Re}\{e^{i\theta+in_0\eta x}p(w_\eta(x))\}.$$

Proof. Substitute the displayed amplitudes and phases. Each Fourier term satisfies $F_t+BF_y=F_{yy}$, and $n_0+q-1\le 2n_0$. □

The coefficients in this last statement may depend on s_*. It therefore cannot replace the fixed-initial-data, full-interval argument of Section 4. Nonvanishing for all $\eta>0$ also says nothing about coefficient size or numerical stability.

C Quantitative consequences and comparison of regimes

This appendix records the dependence of the estimates on an external exponential normalization and compares the static and parabolic scales.

Corollary C.1 (A sufficient coefficient at the growing-mode boundary)**.** *Keep the cutoffs and constants of Theorem 6.2 fixed. Fix $p_0,\gamma>0$, choose $R(L)>0$ so that its geometric hypotheses hold for all sufficiently large L, let $a=p_0L$ and $\omega=\exp(-\gamma L^2)$, and use the normalized quantities*

$$P=R^{-2}\omega M^{\mathrm{plat}}_{\boldsymbol{n},R},\qquad X=(\omega/R)[\mathcal{T}_{\boldsymbol{n},R}]_+.$$

If $q(L)\ge 1$ is integer-valued and $q(L)/L^{5/2}\to\kappa\in[0,\infty)$, the coefficient

$$\mathcal{B}_L=Ca^{2/3}q(L)^7\omega^{1/3}\exp(12\sqrt{2}\,q(L)\sqrt{\Delta_a})$$

in $X\le\mathcal{B}_LP^{2/3}$ satisfies

$$\lim_{L\to\infty}\frac{\log\mathcal{B}_L}{L^2}=-\frac{\gamma}{3}+12\sqrt{2}\,\kappa\sqrt{\frac{\delta+\delta_0}{p_0}}.\tag{67}$$

In particular, the coefficient still decays at a strictly negative quadratic logarithmic rate whenever

$$\kappa < \frac{\gamma}{36\sqrt{2}}\sqrt{\frac{p_0}{\delta+\delta_0}}. \tag{68}$$

The same conclusion holds with a strict limsup in (68).

Proof. Theorem 6.2 gives the normalized inequality by exact cancellation of the powers of R:

$$\frac{\omega}{R}\,a^{2/3}R^{-1/3}(R^2\omega^{-1}P)^{2/3} = a^{2/3}\omega^{1/3}P^{2/3}.$$

Since $\Delta_a = (\delta+\delta_0)/(p_0L-\delta_0)$,

$$\frac{12\sqrt{2}\,q(L)\sqrt{\Delta_a}}{L^2} = 12\sqrt{2}\,\frac{q(L)}{L^{5/2}}\sqrt{\frac{\delta+\delta_0}{p_0-\delta_0/L}}.$$

The limit of this expression is the positive term in (67). The logarithms of $C, a^{2/3}$, and q^7, divided by L^2, tend to zero, and the weight contributes $-\gamma/3$. The same calculation with a limsup proves the last assertion. □

This is a sufficient range for the proved upper-bound coefficient. Neither equality in (68) nor its failure establishes a change in the actual flux. The numerical constants come from the cited extrapolation input. The spatial witnesses in Section B do not supply a matching time-integrated lower bound for this growing-mode boundary.

For the scale choice $p_0 = 32/63$, $\gamma = 2/3969$, and $R = \exp(-9L^2/40000)$, the subcritical rate in Remark 6.3 is $-2/11907$. The exact Chebyshev family of Theorem 4.5 has the same normalized rate in its smaller regime $m = o(A^2)$. These two statements have different strengths: the first is a uniform upper-bound coefficient; the second is an attained rate for the explicitly chosen family.

A power-law parametrization makes the distinction between the static and parabolic scales visible. Let m be the nearest positive even integer to A^σ, with fixed $1/2 < \sigma < 2$. Then

$$\frac{m}{\sqrt{A}} \sim A^{\sigma-1/2}, \qquad \mu \sim A^{(2\sigma-1)/3}, \qquad \mu^2 \sim A^{(4\sigma-2)/3}. \tag{69}$$

The first scale describes the logarithm of a static edge gain. The second describes the edge-dependent exponential factor in the complete-clock quotient, with its algebraic prefactors retained. The third describes the common heat amplitude that cancels from the normalized comparison. In this regime $\mu = o(A)$, and in particular $\mu = o(L^2)$ when A is comparable with L.

The polynomial factors in the flux quotient should not be discarded. For example, $m/\sqrt{A} \to \infty$ allows μ to grow more slowly than $\log A$. In that case an expression described only as $\exp(\Theta(\mu))$ need not imply that the unweighted quotient diverges. The two-sided inequality in Theorem 4.5, rather than that shorthand, is the statement used here.

D A finite diagnostic of the heat-edge calculation

Figure 1 illustrates the three constants in the positive exterior integral used in the proof of Theorem 4.1. Write

$$I_{m,A}(s,c) = \frac{1}{\sqrt{4\pi s}}\int_0^\infty \cosh(m\,\mathrm{arcosh}(1+y/A))\,e^{-(y-c)^2/(4s)}\,dy.$$

At $s = 4$, the normalized saddle location, logarithmic amplitude, and unit-coordinate logarithmic tilt tend respectively to $2^{5/3}$, $3\,2^{-2/3}$, and $2^{-4/3}$. The computation evaluates the positive integral in logarithmic form; it does not obtain a small residual by subtracting large Fourier coefficients.

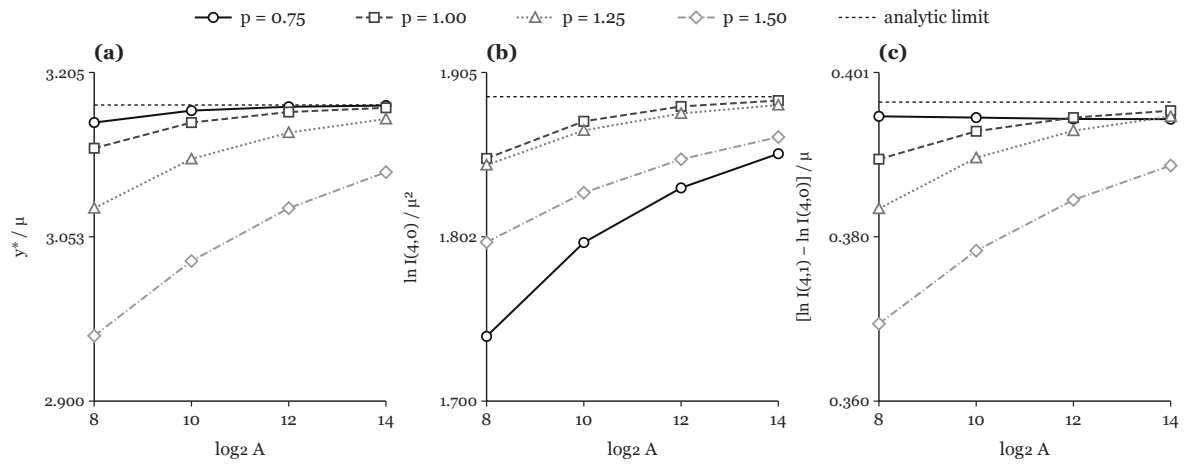


Figure 1: Finite heat-edge diagnostic at $s = 4$. The four degree laws use m equal to the positive even integer nearest A^σ, for $\sigma = 0.75, 1, 1.25, 1.5$, and $A = 256, 1024, 4096, 16384$. Panels (a)–(c) show the saddle location divided by μ, $\log I_{m,A}(4,0)/\mu^2$, and $[\log I_{m,A}(4,1) - \log I_{m,A}(4,0)]/\mu$. Dashed horizontal lines are the three analytic limits. All quantities are dimensionless and natural logarithms are used in (b)–(c). Each panel has a separately enlarged vertical range. The $\sigma = 0.75$ tilt sequence moves slightly away from its limit on this finite grid; convergence is not asserted from these plotted values. The letter p in the figure legend denotes σ here, not the pressure or the scale parameter p_0.

The calculation uses deterministic IEEE-754 binary64 quadrature, with 40 001 grid points for the fine calculation and 20 001 for comparison, and a log-phase drop of 48 to truncate the integration interval. The ancillary directory `anc/heat-edge/` contains the plotting source, parameters, and all 16 data rows; `plot.py -check` reproduces the numerical comparison report. The two-resolution and tail-truncation comparisons measure stability of this finite calculation. They provide no error enclosure for every degree and scale. The limiting assertions and the Navier–Stokes conclusions are justified by the analytic proofs.

This figure illustrates the qualitative limits, not the quantitative bounds or the second-kind extension. Exact algebra checks are included in `anc/checks/`; neither is a continuum certificate.

References

[1] Raed S. Batahan and A. Shehata. Hermite–Chebyshev polynomials with their generalized form. *Journal of Mathematical Sciences: Advances and Applications*, 29:47–59, 2014. Publisher full text.

[2] Tamás Erdélyi. Inequalities for exponential sums. *Sbornik: Mathematics*, 208(3):433–464, 2017. doi:10.1070/SM8670. URL https://arxiv.org/abs/1602.02315v1. Theorem numbers refer to arXiv version 1.

[3] Brian C. Hall and Ching-Wei Ho. The heat flow conjecture for polynomials and random matrices. *Letters in Mathematical Physics*, 115, 2025. doi:10.1007/s11005-025-01946-9. Article 60.

[4] G. N. Hile and Alexander Stanoyevitch. Heat polynomial analogs for higher order evolution equations. *Electronic Journal of Differential Equations*, 2001(28):1–19, 2001. URL https://ejde.math.txstate.edu/Volumes/2001/28/hile.pdf.

[5] Waseem A. Khan. Certain results for the Hermite and Chebyshev polynomials of 2-variables. *Annals of Communications in Mathematics*, 2(2):84–90, 2019. URL https://www.technoskypub.com/wp-content/uploads/2023/09/3-V2N2-P3-W-Certain-results-for-the-Hermite.pdf.

[6] Géza Kós. Two Turán type inequalities. *Acta Mathematica Hungarica*, 119:219–226, 2008. doi:10.1007/s10474-007-6176-5.

[7] National Institute of Standards and Technology. NIST Digital Library of Mathematical Functions, n.d. URL https://dlmf.nist.gov/. Sections 8.4, 16.2, 16.11 and 18.5; accessed 6 September 2026.

[8] R. B. Paris. Asymptotics of a Gauss hypergeometric function with large parameters, IV: A uniform expansion, 2018. URL https://arxiv.org/abs/1809.08794v2. Version 2, 14 October 2018.

[9] R. B. Paris. Uniform asymptotics of a Gauss hypergeometric function with two large parameters, V, 2021. URL https://arxiv.org/abs/2004.01945v2. Version 2, 25 April 2021; first submitted 2020.

[10] Ruizhe Zhang. Optimal extrapolation bounds for sparse Fourier sums, 2026. URL https://arxiv.org/abs/2607.10501v1. Version 1, 11 July 2026; preprint.